

μ 's Paper 3

Time allowed: 1 hour 15 minutes

This is a practice Paper 3 for all AA HL students. It is probably harder than what will come up on your final examination, but nevertheless it should be good practice. Enjoy!

Instructions

- Answer all questions.
- Show all necessary working clearly.
- Unless otherwise stated, exact values are preferred.
- A graphic display calculator is required for this paper.

May 2026

1. [Maximum mark: 29]

The following question investigates a family of curves defined by a constant product of distances from two fixed points.

Consider two fixed points $A(-a, 0)$ and $B(a, 0)$, where $a \in \mathbb{R}^+$.

A point $P(x, y)$ moves so that the product of its distances from A and B is constant:

$$|\overrightarrow{PA}| \cdot |\overrightarrow{PB}| = b^2,$$

where $b > 0$.

The curve traced out by P is called a *Cassini oval*. This family of curves has many applications in physics as it is linked to the lines of equipotential around massive bodies.

(a) (i) Write the equation of the curve traced by P . [2]

(ii) Hence show that

$$((x + a)^2 + y^2)((x - a)^2 + y^2) = b^4. \quad [1]$$

(b) (i) By expanding the left-hand side, show that

$$((x + a)^2 + y^2)((x - a)^2 + y^2) = (x^2 + y^2 + a^2)^2 - 4a^2x^2.$$

(ii) Hence deduce that the equation of the curve can be written as

$$(x^2 + y^2 + a^2)^2 - 4a^2x^2 = b^4. \quad [4]$$

(c) Using the equation from (b) (ii) explain why the curve is symmetric about both the x -axis and the y -axis. [2]

(This question continues on the following page.)

- (d) (i) Show that the x -intercepts satisfy

$$|x^2 - a^2| = b^2.$$

(ii) Hence find all possible x -intercepts.

(iii) Show that the intersections with the y -axis exist only when

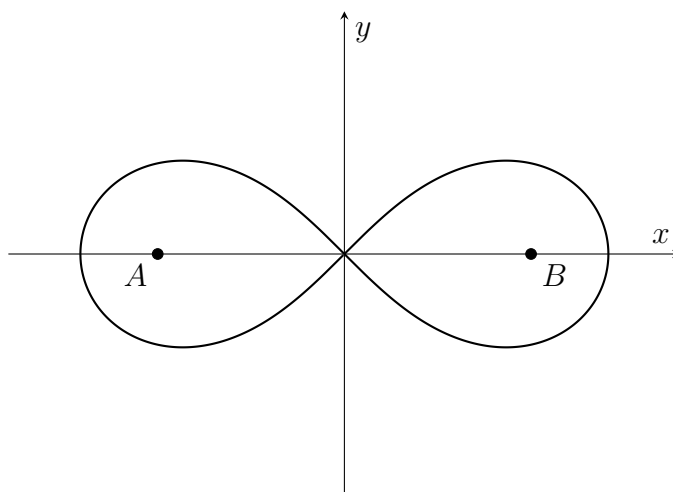
$$b \geq a. \quad [6]$$

- (e) Now consider the case $b = a$.

Show that the equation of the curve becomes

$$(x^2 + y^2)^2 = 2a^2(x^2 - y^2). \quad [4]$$

This curve is called the *lemniscate of Bernoulli*.



Sketch of the lemniscate of Bernoulli

- (f) Using

$$x = r \cos \theta, \quad y = r \sin \theta,$$

show that the curve can be written

$$r^2 = 2a^2 \cos(2\theta). \quad [3]$$

- (g) For the right-hand loop of the lemniscate,

$$-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}.$$

By writing the expression for the curve in (f) as $y^2 = f(\theta)$, where $f(\theta)$ is a function to be determined, or otherwise, find the maximum value of y on the right-hand loop. [7]

2. [Maximum mark: 26]

This question investigates a sequence of nested radicals and shows how it leads to a general infinite product formula.

Consider the sequence (a_n) defined by

$$a_0 = 0, \quad a_{n+1} = \sqrt{2 + a_n}, \quad n \geq 0.$$

(a) Find a_1 , a_2 and a_3 , giving your answers in exact form. [2]

(b) (i) Show that, for $0 \leq \theta \leq \pi$,

$$\sqrt{2 + 2 \cos \theta} = 2 \cos \left(\frac{\theta}{2} \right). \quad [2]$$

(ii) Hence prove the following statement by induction for all integers $n \geq 0$:

$$a_n = 2 \cos \left(\frac{\pi}{2^{n+1}} \right).$$

[6]

(c) We now define the sequence $(P_n)_{n \in \mathbb{N}}$ as follows:

$$P_n = \prod_{k=1}^n \frac{a_k}{2},$$

where $\prod$ denotes product notation:

$$\prod_{i=1}^n u_i = u_1 \cdot u_2 \cdot u_3 \cdots u_n.$$

(i) Show that

$$P_n = \prod_{k=1}^n \cos \left(\frac{\pi}{2^{k+1}} \right). \quad [1]$$

(ii) By using the identity

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

repeatedly or otherwise, show that

$$1 = 2^n \sin \left(\frac{\pi}{2^{n+1}} \right) P_n. \quad [5]$$

(This question continues on the following page.)

(iii) Hence show that

$$P_n = \frac{1}{2^n \sin\left(\frac{\pi}{2^{n+1}}\right)}.$$

Using the standard limit

$$\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1,$$

or otherwise, deduce that

$$\lim_{n \rightarrow \infty} P_n = \frac{2}{\pi}. \quad [3]$$

This gives the infinite product

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2 + \sqrt{2}}}{2} \cdot \frac{\sqrt{2 + \sqrt{2 + \sqrt{2}}}}{2} \dots$$

(d) The result above can be generalised.

For $0 < x < \pi$, we define a new sequence $(b_n(x))$ by

$$b_0(x) = 2 \cos x, \quad b_{n+1}(x) = \sqrt{2 + b_n(x)}.$$

You are given that

$$b_n(x) = 2 \cos\left(\frac{x}{2^n}\right)$$

for all integers $n \geq 0$.

We now introduce

$$Q_n(x) = \prod_{k=1}^n \frac{b_k(x)}{2}.$$

(i) Show that

$$Q_n(x) = \frac{\sin x}{2^n \sin\left(\frac{x}{2^n}\right)}. \quad [4]$$

(ii) Hence deduce that

$$\prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right) = \frac{\sin x}{x}$$

for $0 < x < \pi$.

[3]